

Shortest paths in DAGs

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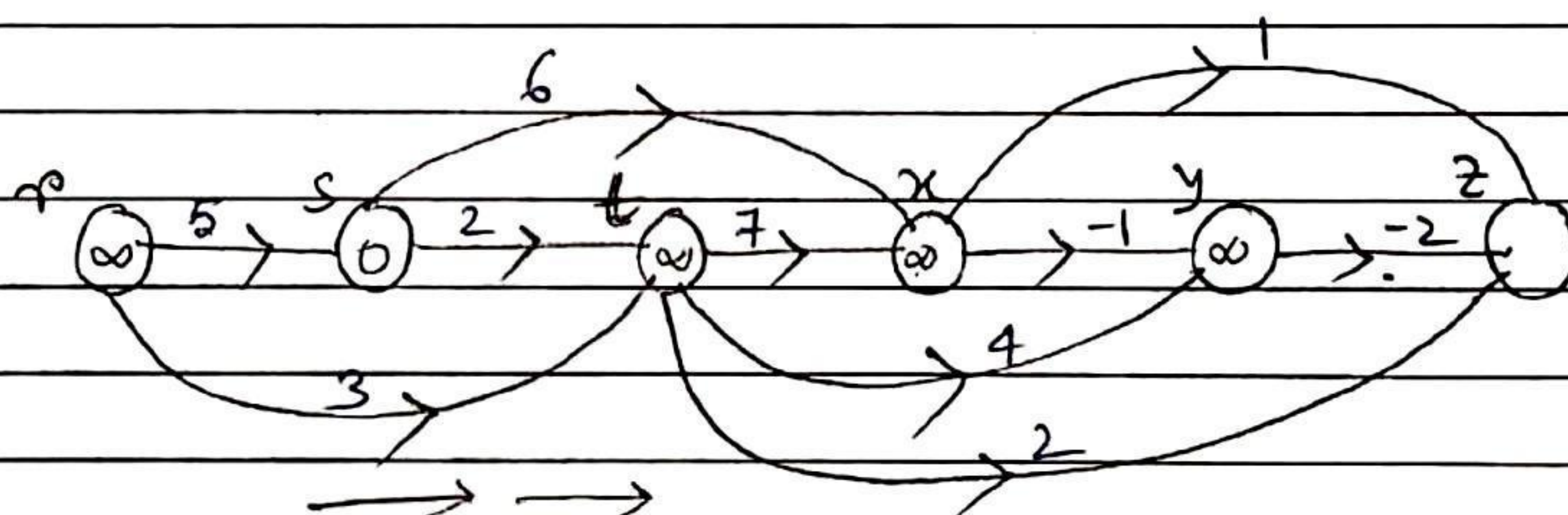
- Shortest paths in DAG (Directed acyclic graph) =

→ $O(V+E)$ time take to put a DAG in topological order. (Edges right to left)

→ Take the vertices one by one topological order and then try to relax ~~edges~~ outgoing edges outgoing from them.

example: 1

What is the shortest path from S to all the other vertices.



	r	s	t	u	y	z
r	∞	0	∞	∞	∞	∞
s	∞		2	6	∞	∞
t	∞			6	6	4
u	∞				5	4
y	∞					3

Topological sort = $O(V+E)$

Total relaxation done = $O(E)$.

Total time complexity = $O(V+E)$

DAG - shortest - paths (G, w, s)

- $$\{$$
- 1) Topologically sort the vertices of G . — $O(V+E)$ order
 - 2) Initialize single-source (G, s) . — $O(V)$ sorted
 - 3) for each vertex u , taken in topologically order
 - 4) for each vertex $v \in G.\text{adj}[u]$
 - 5) Relax(u, v, w)
- $$\} \quad \text{--- } O(E)$$
- $$\}$$

Time complexity = $O(V+E)$.